

Introduction to Programming



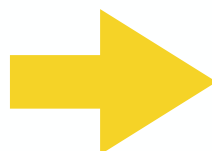
W02 Control Flow in Programming

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Schedule



#	Date	Subject
1	16.10.24	Introduction
2	23.10.24	Control Flow in Programming
3	30.10.24	In-Depth Core Concepts*
4	06.11.24	Core Data Structures
5	13.11.24	Code Reuse and Structure
6	20.11.24	Type Flexibility and Safety
7	27.11.24	In-Depth Object Orientation*
8	04.12.24	Functional Programming Essentials
9	11.12.24	Algorithms and Data Handling
10	18.12.24	Programming Languages
11	08.01.25	Graphical User Interfaces
12	15.01.25	Recursion
13	22.01.25	Concurrency
14	29.01.25	Beyond Programming
15	05.02.25	Course Review

* Repetition

- **Context**

- Understand the difference between objects and classes
- Implement attributes, constructors and methods
- Instantiate objects using constructors and invoke methods

- **Learning goals**

- Implement conditional statements using **if** and **switch**
- Properly deal with **null** values
- Implement iterations using **for**, **while** and **do-while**
- Use **arrays** to store multiple elements of the same type
- Implement basic operations such as **search** and **sort**

Outline

Expressions and statements

- Conditional selection
- Iteration
- Arrays
- Search
- Sort

Motivation



- A programming language should
 - Provide data structures
 - Allow operations on data
 - Provide control structures for flow control
- We have already looked first at data structures and operations
- Control structures use **statements** and **expressions**

Expressions

- Appear in a program...
 - ... on the right side of assignments
 - ... as arguments of methods
 - ... in the body of functions
- Each expression has a **type**

```
x = expression;
```

```
f(expression1, ..., expressionn);
```

```
return expression;
```

- **Examples**

- Expression of type **double**

```
(3.0 + y) * 4.0
```

- Expression of type **String**

```
"Hello" + " " + "World" + 2
```

- Expression of type **Point**

```
new Point(1.0, 2.0)
```

- Expression of type **boolean**

```
x == 2
```

Also called condition

Expressions

- Expressions are composed as follows
 - `Variable` or `Attribute`
 - `f(expression1, ..., expressionn)` (if return type is **not void**)
 - `new ...`
 - `Expression1 ⊕ Expression2`

Binary operator, e.g. `a + b` or `c - d`
 - `⊙ expression`

Unary operator, e.g. `- fahrenheit(15)`
- Two other expressions in Java
 - `_ instanceof _` class membership test
 - `_ ? _ : _` conditional expression (if-then-else, ternary operator)

Statements

- Typically cause a change of state
- End with a semicolon (;) and do **not** have a type
- In Java, there are a variety of statements

- Declaration of local variables

```
int x;
```

- Assignment

```
x = y;
```

- Method call
(if return type is **void**)

```
f(x, 1);
```

- Block statement

```
{ statement1; ... ; statementn; }
```

- Return statement

```
return expression;
```


Comparison expression - statement

	Expression	Statement
Typed	Yes	No
Purpose	Calculation	Execution
Effect	Evaluates a value	Changes the state
Syntax	Without ;	With ;

- Statements may contain expressions
- The body of each method is always a sequence of statements
- In Java, there are also so-called expression statements
 - These are statements that can also be used as an expression at the same time
 - **Example:** `i++;`

Sequence of statements

Example

```
int x, y, result;  
x = InputReader.readInt("Number 1: ");  
y = InputReader.readInt("Number 2: ");  
result = x + y;  
System.out.println("Sum: " + result);
```

Note: please copy the **InputReader** on <https://gist.github.com/krusche/f8bdf092159cc272f5e3ff513f2b1cb5>

Sequence of statements

- Only **one operation** is performed at any time
 - Each operation is performed exactly once one after the other
 - None is repeated, none omitted
 - Order of execution as defined in the program code
 - At the end of the last operation, the program execution ends
- A sequence of statements only allows very simple programs
- **We need more powerful control structures**